

Question 8

i) On the y -axis, $f(x, y) = f(0, y) = 0$.

Hence, $\lim_{y \rightarrow 0} f(0, y) = 0$.

On the curve $x = y^2$, $f(x, y) = \frac{y^4}{2y^4} = \frac{1}{2}$.

Hence, $\lim_{y \rightarrow 0} f(y^2, y) = \frac{1}{2}$.

Since these two limits are different,

$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^2}$ doesn't exist.

ii) For any point (x, y) , let r be its distance to the origin.

Then $r^2 = x^2 + y^2$,

$|x| \leq r$ and $|y| \leq r$.

Hence, $\left| \frac{3x^2y}{x^2+y^2} \right| \leq \frac{3r^3}{r^2} = 3r$

$\Rightarrow \left| \frac{3x^2y}{x^2+y^2} \right| \rightarrow 0$ as $r \rightarrow 0$

(i.e. as $(x, y) \rightarrow (0, 0)$)

$\rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2+y^2} = 0$

Question 9

i) $f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$

$= \lim_{h \rightarrow 0} \frac{(x+h)y^2 - (x+h)^2 y - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} \frac{xy^2 + hy^2 - x^2 y - 2xhy - h^2 y - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} y^2 - 2xy - hy = y^2 - 2xy$

ii) $f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}$

$= \lim_{h \rightarrow 0} \frac{x(y+h)^2 - x^2(y+h) - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} \frac{xy^2 + 2xyh + xh^2 - x^2 y - x^2 h - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} 2xy + xh - x^2 = 2xy - x^2$

Question 9

$$i) f_x(x, y) = \lim_{h \rightarrow 0} \frac{\frac{x+h}{x+h+y^2} - \frac{x}{x+y^2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)(x+y^2) - x(x+h+y^2)}{(x+h+y^2)(x+y^2)h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + xy^2 + xh + y^2h - x^2 - xh - xy^2}{(x+h+y^2)(x+y^2)h}$$

$$= \lim_{h \rightarrow 0} \frac{y^2}{(x+h+y^2)(x+y^2)} = \frac{y^2}{(x+y^2)^2}$$

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{\frac{x}{x+(y+h)^2} - \frac{x}{x+y^2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x(x+y^2) - x(x+(y+h)^2)}{(x+(y+h)^2)(x+y^2)h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + xy^2 - x^2 - x(y^2 + 2yh + h^2)}{(x+(y+h)^2)(x+y^2)h}$$

$$= \lim_{h \rightarrow 0} \frac{-2xy - xh}{(x+(y+h)^2)(x+y^2)} = \frac{-2xy}{(x+y^2)^2}$$

Question 10

$$i) \frac{\partial}{\partial x} (yz) = \frac{\partial}{\partial x} \ln(x+z)$$

$$y \frac{\partial z}{\partial x} = \frac{1}{x+z} \left(1 + \frac{\partial z}{\partial x}\right)$$

$$\left(y - \frac{1}{x+z}\right) \frac{\partial z}{\partial x} = \frac{1}{x+z}$$

$$\frac{\partial z}{\partial x} = \frac{\frac{1}{x+z}}{y - \frac{1}{x+z}} = \frac{1}{y(x+z) - 1}$$

$$\frac{\partial}{\partial y} (yz) = \frac{\partial}{\partial y} \ln(x+z)$$

$$z + y \frac{\partial z}{\partial y} = \frac{1}{x+z} \frac{\partial z}{\partial y}$$

$$\left(y - \frac{1}{x+z}\right) \frac{\partial z}{\partial y} = -z$$

$$\frac{\partial z}{\partial y} = \frac{-z}{y - \frac{1}{x+z}} = \frac{-z(x+z)}{y(x+z) - 1}$$

$$ii) \frac{\partial}{\partial x} (x - z) = \frac{\partial}{\partial x} \arctan(yz)$$

$$1 - \frac{\partial z}{\partial x} = \frac{1}{1+(yz)^2} \cdot y \frac{\partial z}{\partial x}$$

$$\left(1 + \frac{y}{1+(yz)^2}\right) \frac{\partial z}{\partial x} = 1$$

$$\frac{\partial z}{\partial x} = \frac{1}{1 + \frac{y}{1+(yz)^2}} = \frac{1+(yz)^2}{1+(yz)^2 + y}$$

$$\frac{\partial}{\partial y} (x - z) = \frac{\partial}{\partial y} \arctan(yz)$$

$$-\frac{\partial z}{\partial y} = \frac{1}{1+(yz)^2} \left(z + y \frac{\partial z}{\partial y}\right)$$

$$\left(1 + \frac{y}{1+(yz)^2}\right) \frac{\partial z}{\partial y} = \frac{-z}{1+(yz)^2}$$

$$\frac{\partial z}{\partial y} = \frac{\frac{-z}{1+(yz)^2}}{1 + \frac{y}{1+(yz)^2}} = \frac{-z}{1+(yz)^2 + y}$$

Question 11

$$i) f_x(x, y) = \sin(x+zy) + x \cos(x+zy)$$

$$f_{xy}(x, y) = z \cos(x+zy) - zx \sin(x+zy)$$

$$f_y(x, y) = zx \cos(x+zy)$$

$$f_{yx}(x, y) = z \cos(x+zy) - zx \sin(x+zy)$$

Same

$$ii) f_x(x, y) = 4x^3 y^2 - 2y^5$$

$$f_{xy}(x, y) = 8x^3 y - 10y^4$$

$$f_y(x, y) = 2x^4 y - 10xy^4$$

$$f_{yx}(x, y) = 8x^3 y - 10y^4$$

Same

Question 12

$$i) f_x(x, y) = \frac{3}{3x+5y}$$

$$f_y(x, y) = \frac{5}{3x+5y}$$

$$f_{xx}(x, y) = \frac{-3}{(3x+5y)^2}$$

$$f_{xy}(x, y) = f_{yx}(x, y) = \frac{-15}{(3x+5y)^2}$$

$$f_{yy}(x, y) = \frac{-25}{(3x+5y)^2}$$

$$i) \quad z_x = \frac{x+y-x}{(x+y)^2} = \frac{y}{(x+y)^2}$$

$$z_y = \frac{-x}{(x+y)^2}$$

$$z_{xx} = \frac{-2y}{(x+y)^3}$$

$$z_{xy} = z_{yx} = \frac{(x+y)^2 - 2y(x+y)}{(x+y)^4} = \frac{x+y-2y}{(x+y)^3} = \frac{x-y}{(x+y)^3}$$

$$z_{yy} = \frac{2x}{(x+y)^3}$$

Question 13

$$i) \quad f_t = -cx^2 e^{-ct}$$

$$f_{tt} = c^2 x^2 e^{-ct}$$

$$f_{ttt} = -c^3 x^2 e^{-ct}$$

$$f_{tx} = -2cx e^{-ct}$$

$$f_{txx} = -2c e^{-ct}$$

$$ii) \quad f_x = -4 \sin(4x+3y+2z)$$

$$f_{xy} = -12 \cos(4x+3y+2z)$$

$$f_{xyz} = 24 \sin(4x+3y+2z)$$

$$= f_{zyx}$$

Question 14

$$i) \quad \text{The ellipse is given by } 4x^2 + 8 + z^2 = 16$$

$$\Rightarrow 4x^2 + z^2 = 8$$

To find the slope of the tangent line, we use implicit differentiation:

$$8x + 2z \frac{dz}{dx} = 0$$

$$\frac{dz}{dx} = -\frac{8x}{2z} = -4 \frac{x}{z}$$

$\Rightarrow$ The tangent line at the point $(1, 2, 2)$ has slope $-4 \cdot \frac{1}{2} = -2$.

$\Rightarrow$ The equation of the tangent line is

$$z - 2 = -2(x - 1)$$

$$z = -2x + 4$$